

Rocky Chang | Curriculum Vitae

Department of Computer Science, Calvin University

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Professional Experience

Calvin University, Grand Rapids, MI

Professor

2023–Present

Department of Computer Science

Chung Yuan Christian University (CYCU), Taiwan

Associate Professor

2022–2023

Department of Information and Computer Engineering

CYCU, Taiwan

Visiting Associate Professor

2020–2022

Department of Information and Computer Engineering

University of Technology Sydney, Australia

Key Technology Partner Visiting Fellow

2016

Hong Kong Polytechnic University (PolyU)

Associate Professor

2002–2020

Department of Computing

PolyU

Assistant Professor

1993–2002

Department of Computing

IBM T.J. Watson Research Center, NY

Advisory Programmer

1991–1993

Education

Rensselaer Polytechnic Institute (RPI), NY

PhD in Computer and Systems Engineering

1990

RPI, NY

MSc in Operations Research and Statistics

1987

RPI, NY

MEngr in Electrical Engineering

1985

Virginia Tech, VA

BSc in Electrical Engineering

1983

Teaching

Leadership

2025: Designed a certificate program on cybersecurity, CS, Calvin

2024–Present: Chairman, Curriculum Committee, CS, Calvin

2015–2019: Chairman, Undergraduate Program Committee, Computing, PolyU
1996–1997: Chair, Revalidation Committee, BSc (Hons) in Information Technology, Computing, PolyU
1995–1996: Founding Award Coordinator, MSc in Software Technology, Computing, PolyU
1994–1995: Chair, Program Planning Committee, MSc in Software Technology, Computing, PolyU
1993–1996: Founding Course Leader, BSc (Hons) in Information Technology, Computing, PolyU

Awards

2001/2002: First Runner-up, Faculty of Engineering Best Teacher Award, PolyU

Extra-Curricular Activities (as a founding member)

2016–2017: PolyHack – First hackathon at PolyU
2014–2018: Computing Solved It! (CSI!) Competition for PolyU's Computing Freshmen
2014: Python Programming Contest, Computing, PolyU
2014–2015: Summer workshops on C, Python, Android, Raspberry Pi, GO, Linux, drones, etc

Grants by PolyU

1995–1998: A Pilot Study of Self-Paced Subject Learning Based on Hypermedia Environment (HKD \$237,680)
1995–1997: Learning Computer Communication Networks Through Software-Based Experiments (HKD \$98,000)
1995: Design and Development of Laboratory Materials on Networking Software (HKD \$89,160)

Seminars

2019: *"How to Engage Students' Minds in Our Teaching?"* PolyU
2015: *"What Can and Should We Do About Academic Dishonesty?"* PolyU
2014: *"On Our 16-week Teaching and Learning Experience in COMP1001,"* PolyU
2013: Panel on *"Experience Sharing on Deepening Computer Network Teaching with Network-based Information Platform,"* Sixth National Workshop on Teaching Computer Networks, Huhhot, PRC
2013: *"My Research Students and I,"* National University of Defense Technology, Changsha, PRC
2012: *"My Reflections on What to Teach in Computer Networking,"* Keynote speech, Fifth National Workshop on Teaching Computer Networks, Wenzhou, PRC
2009: *"WIE@COMP: Managing the Diversity of WIE Activities,"* PolyU
2005: *"Active Learning Experience in a Year-long Placement Programme,"* PolyU
2004: *"Believing in Our Teaching and Our Students,"* PolyU
2004: *"The Role of Student Feedback in Active Learning,"* PolyU
2004: *"Architecting an Active Classroom: An Integrative Approach,"* PolyU
2003: *"Open Book Examinations: What It Is and What It Is Not,"* PolyU
2003: Panel Session on *"What Works and What Doesn't Work: Tips for New Networking Educators,"* ACM Workshop on Networking Education, Germany
2003: *"Trading Teaching for Research? A Personal Testimony,"* PolyU
2003: *"Assessing the Impact of Our Teaching,"* PolyU
2002: *"Transparent Teaching,"* PolyU

2001: "Sharing of Some Personal Thoughts on Teaching," PolyU

Publications

- [1] Rocky K. C. Chang. "Using Problem-Driven Learning to Teach Computational Thinking in CS1." In: *Proc. Consortium for Computing Sciences and Colleges: Midwest Conf.* Sept. 2024.
- [2] Rocky K. C. Chang. "Teaching Computer Networking with the Help of Personal Computer Networks." In: *Proc. 9th Annual ACM Conf. Innovation and Technology in Computer Science Education.* June 2004.
- [3] Rocky K. C. Chang. "An Integrated View of Teaching and Learning for a Foundational Course on Computer Networking." In: *Proc. ACM Workshop on Networking Education.* Aug. 25, 2003.
- [4] Rocky K. C. Chang and Man K. Lee. "Learning Computer Communication Network Through Software-based Experiments." In: *Case Studies of Improving Teaching and Learning from the Action Learning Project.* D. Kember et al., 1997, pp. 429–438.

Other Write-ups

2025: "Cybersecurity: Digital Stewardship in a Violent World," Calvin

2024: "Living the Gospel in Computer Science Teaching," Calvin

2003: "My Beliefs in Teaching and Students," PolyU

2001: "My Teaching Philosophy, Approaches, and Effectiveness," PolyU

Course Development

2025: Designed a new elective "Network Security," Calvin

2024: Introduced a new set of labs and weekly code reviews to "Introduction to Computing" (CS108), Calvin

2022: Introduced a new set of labs on Git, Github and GitPi and MQTT to "Embedded Systems" (CS014), CYCU

2021: Introduced individual presentations to "English Presentation Skills" (CS222G), CYCU

2021: Introduced five new modules to "Engineering Ethics" (GE488), CYCU

2020: Designed a new general education course "Guided Exploration of the Internet" (GE733), CYCU

2021: Designed a new course "Computational Thinking and Program Design" (CS888), CYCU

2021: Authored a syllabus for "Introduction to EE and CS" (EE101) for the Wisconsin-CYCU joint program, CYCU

2014–2018: Designed a new course "Problem-Solving Methodology in Information Technology" (COMP1001), PolyU

2018: Developed a lecture-lab integrated teaching approach for "Computer Networking" (COMP2322), PolyU

2012: Designed two new cybersecurity courses: Computer Systems Security (COMP3334) and Principles and Practice for Internet Security (COMP4334), PolyU

2006: Designed a new final-year elective "Internet Infrastructure Security" (COMP444), PolyU

1998: Designed a new MSc course "Internetworking Protocols and Software II" (COMP555), a sequel to COMP526, PolyU

1998: Updated the syllabus of "Selected Areas in Computer Networks" (COMP530) to more

up-to-date topics, such as ATM technology, IP over ATM networks, and label switching networks, PolyU

1996: Promptly re-aligned the focus of an undergraduate course on "Computer Communication Networks" (COMP312) to an Internet-centric approach, PolyU

1993: Designed a new MSc course "Internetworking Protocols and Software" (COMP526, COMP5311), PolyU

Subjects Taught

2023-2025: Calvin: Introduction to Computing (CS108), and Operating Systems and Networking (CS232)

2020-2023: CYCU: Computational Thinking and Program Design, Introduction to Embedded Systems, Internet Infrastructure Security, Principles of Internetworking Protocols, Advanced Internet Infrastructure, Engineering Ethics, Guided Exploration of the Internet, and English Presentation Skills

1993-2019: PolyU: Problem-Solving Methodology in Information Technology, Computer Programming, Computer Organization, Data Structure and Algorithms, Computer Networking, Internetworking Protocols and Software, Internet Infrastructure Security, and Selected Topics in Computer Network

Research

Interests

Internet infrastructure and security with specialization on network performance measurement and monitoring, quality-of-experience measurement. denial-of-service attacks and defenses, Android security, performance modeling, CS education, and CS faith integration

Leadership

2024–Present: Member, STEM Research Committee, Calvin

2016–2017: Founding Director, Cyber Security and Privacy Lab, Computing, PolyU

2015–2020: Leader, a research group on Internet Services Monitoring and Diagnostics in the Division of Smart Cities under PolyU's Research Institute for Sustainable Urban Development, PolyU

2003–2020: Leader, Internet Infrastructure and Security Research Group, Computing, PolyU

2012: Founding Coordinator, Annual Research Day (CARD), Computing, PolyU

2010–2012: Member, Research Committee, Computing, PolyU

1996–1997: Member, Research Committee, Faculty of Business and Information System, PolyU

1993–1996: Member, Research Committee, Computing, PolyU

Awards

2015: Most Active New Consultant Award (Merit), PolyU

2009, 2012: Industrial and Engineering Services Grant Achievement Award, Faculty of Engineering, PolyU

2007: Best Student Paper Award (Xiapu Luo and Edmond Chan) in the 22nd TC-11 IFIP Intl. Information Security Conf. (May 2007) for "Crafting Web Counters into Covert Channels"

2007: Honorable Mention Award (Yi Xie) in the BJ-HK International Doctoral Forum for "Optimizing Energy Efficiency in Polling-Based Wireless Networks with Stability Constraints"

Funded Projects (As Chief Investigator).....

Externally Funded

2025–2028: Collaborative Research: CyberTraining: Implementation: Medium: Engaging Scholars in Cybersecurity Analysis: A Laboratory for Teaching and Education (ESCA-LATE). Funded by NSF (Training-based Workforce Development for Advanced Cyberinfrastructure), collaborators: UCSD and Johns Hopkins, total project cost: USD \$163,803.

2020–2021: Trial: A Platform for Residential Broadband Network Monitoring. Funded by the Hong Kong Innovation and Technology Commission (Public Sector Trial Scheme), total project cost: HKD \$1,014,000.

2016: Research on OTT Traffic Modeling and its Impact on Network Architecture. Funded by Huawei's HIRP, total project cost: HKD \$615,000.

2012–2013: A Novel System for Assessing the Security of Android Apps. Funded by the Hong Kong Innovation and Technology Commission (ITSP tier-three programme), total project cost: HKD \$997,625.

2012–2014: Design and Implementation of a Unified Box for Offering Network Path Measurement as a Service. Funded by the HK Innovation and Technology Commission (Guangdong-Hong Kong Technology Cooperation Scheme), total project cost: HKD \$4,963,700.

2012–2014: A Novel System for Assessing the Security of Android Apps. Funded by the HK Innovation and Technology Commission (ITSP tier-three programme), total project cost: HKD \$997,625.

2010–2011: Reliable and Accurate Bandwidth Measurement of Asymmetric Network Paths. Funded by the HK Innovation and Technology Commission (ITSP tier-three programme), total project cost: HKD \$967,150.

2008–2010: Uncooperative Measurement and Monitoring of Internet Path Quality with Applications. Funded by the HK Innovation and Technology Commission (ITSP tier-three programme), total project cost: HKD \$999,350.

2005–2006: Analyzing the Impact of Smart Denial-of-Service Attacks on TCP and Queue Management Scheme. Funded by Cisco CCRP, total project cost: USD \$4,500.

2004–2005: Area of Excellence in Information Technology (Network Security). Funded by the University Grants Council of Hong Kong, total project cost: HKD \$900,000.

2002–2005: A Distributed Detection Approach to Flooding-based Distributed Denial-of-service Attack Prevention. Funded by the Competitive Earmarked Research Grant, total project cost: HKD \$440,000.

2001–2004: Toward a General Approach to Per-queue Stability Problems with Applications in Wireless Communications. Funded by the Competitive Earmarked Research Grant, total project cost: HKD \$400,000.

Internally Funded

2025–2026: Crowdsourcing Internet Speed Measurement from Multiple Speed Test Platforms. Funded by Calvin Research Fellowship and STEM summer research.

2014–2016: Practical Monitoring and Troubleshooting Cross-domain Network Paths with a Single Monitor. Funded by the PolyU Central Research Grant, total project cost: HKD \$150,000.

2011–2013: Understanding the Global Internet's Vulnerability to Submarine Cable Faults.

Funded by the PolyU Central Research Grant, total project cost: HKD \$105,000.

2010–2012: Modeling the False Feedback Injection Attacks with Switched Systems.

Funded by the PolyU Central Research Grant, total project cost: HKD \$105,000.

2007–2009: Is Covert Timing Communication a Real Threat to the Internet? Funded by the PolyU Central Research Grant, total project cost: HKD \$150,000.

2004–2006: BGP-based Inbound Traffic Engineering for Multihomed Autonomous Systems: A Measurement-based Analysis. Funded by the PolyU Central Research Grant, total project cost: HKD \$150,000.

2003–2006: Early Detection of Flash Crowds: A Nonparametric Approach. Funded by the PolyU Central Research Grant, total project cost: HKD \$170,000.

2000–2004: Multicast Tree Aggregation in MPLS Networks: Protocol Design and Performance Analysis. Departmental General Research Fund, total project cost: HKD \$450,000.

1998–2002: Deadlock-free Silly Window Syndrome Avoidance Algorithms for High-speed Internet. Funded by the PolyU Central Research Grant, total project cost: HKD \$240,000.

1997–1999: An Empirical Study of the Impact of Self-Similar Traffic Sources on the Stability of Multiaccess Networks. Departmental General Research Fund, total project cost: HKD \$31,000.

1996–1998: The Study of High-speed LAN based on MPP Technology. Funded by the PolyU Central Research Grant, total project cost: HKD \$186,000.

1994–1997: Capacity Modelling of Individual Hosts on Multi-access Networks. Departmental General Research Fund, total project cost: HKD \$467,000.

Consultancy

2009–2019: Performance Monitoring and Measurement of HARNET, funded by the Joint Universities Computer Centre, HK (total project cost: over HKD 2 million).

2013–2019: Performance Monitoring of Critical Network and Service Infrastructure in Hong Kong, funded by a HK government department (total project cost: over HKD 1.5 million).

US Patent

2013: US Patent US8531952 B2—Method for measurement of network path capacity with path capacity discovery (PCD) (with Edmond Chan and Xiapu Luo)

Working Papers

2025: "Demystifying the Ookla Speed Test Measurement" (with Ben Kusters and Ricky Mok)

2025: "Tracking the Evolution of Geofeeds: Data Validation and Adoption" (with Daniel Park, Hye Chan, and Ricky Mok)

2025: "CARROT: Configurable and Reproducible Client for Open Speed Test" (with Hemil Panchiwala, Anwesha Pradhananga, Aishwarya Joshi, and Ricky Mok)

Refereed Publications

[1] Daoyuan Wu, Debin Gao, Robert H. Deng, and Rocky K. C. Chang. "When Program Analysis Meets Bytecode Search: Targeted and Efficient Inter-procedural Analysis of Modern Android Apps in BackDroid." In: *Proc. DSN*. 2021.

[2] Shiwei Zhang, Weichao Li, Daoyuan Wu, Bo Jin, Rocky K. C. Chang, Debin Gao, Yi Wang, and Ricky K. P. Mok. "An Empirical Study of Mobile Network Behavior and Application

- Performance in the Wild." In: *Proc. IEEE/ACM International Symposium on Quality of Service (IWQoS)*. 2019.
- [3] Daoyuan Wu, Debin Gao, Rocky K. C. Chang, En He, Eric K. T. Cheng, and Robert H. Deng. "Understanding Open Ports in Android Applications: Discovery, Diagnosis, and Security Assessment." In: *Proc. ISOC Network and Distributed System Security Symposium (NDSS)*. 2019.
 - [4] Steven Lee and Rocky K. C. Chang. "Enhancing the Experience of Multiplayer Shooter Games via Advanced Lag Compensation." In: *Proc. ACM MMSys*. June 2018.
 - [5] Weichao Li, Daoyuan Wu, Rocky K. C. Chang, and Ricky Mok. "Toward Accurate Network Delay Measurement on Android Phones." In: *IEEE Trans. Mobile Computing* (Aug. 2017).
 - [6] Daoyuan Wu, Rocky K. C. Chang, Weichao Li, Eric K. T. Cheng, and Debin Gao. "MopEye: Opportunistic Monitoring of Per-app Mobile Network Performance." In: *Proc. USENIX Annual Technical Conference*. July 2017.
 - [7] Steven Lee and Rocky K. C. Chang. "On 'Shot Around a Corner' in First-Person Shooter Games." In: *Proc. NetGames*. June 2017.
 - [8] Ricky Mok, Rocky K. C. Chang, and Weichao Li. "Detecting Low-quality Workers in QoE Crowdfunding: A Worker Behavior Based Approach." In: *IEEE Trans. Multimedia* (Oct. 2016).
 - [9] Ricky K. P. Mok, Weichao Li, and Rocky K. C. Chang. "IRate: Initial Video Bitrate Selection System for HTTP Streaming." In: *IEEE J. Sel. Areas in Communications (SI on Measuring and Troubleshooting the Internet: Algorithms, Tools and Applications)* (June 2016).
 - [10] Weichao Li, Daoyuan Wu, Rocky K. C. Chang, and Ricky Mok. "Demystifying and Puncturing the Inflated Delay in Smartphone-based WiFi Network Measurement." In: *Proc. ACM CoNEXT (short paper)*. Dec. 2016.
 - [11] Peter Membrey, Darryl Veitch, and Rocky K. C. Chang. "Time to Measure the Pi." In: *Proc. ACM IMC (short paper)*. Oct. 2016.
 - [12] Peng Zhou, Rocky K. C. Chang, Xiaojing Gu, Minrui Fei, and Jianying Zhou. "Magic Train: Design of Measurement Methods Against Bandwidth Inflation Attacks." In: *IEEE Trans. Dependable and Secure Computing* (Dec. 2015).
 - [13] Peng Zhou, Xiaojing Gu, and Rocky K. C. Chang. "Harvesting File Download Exploits in the Web: A Hacker's View." In: *The Computer Journal* (Sept. 2015).
 - [14] Ricky K. P. Mok, Weichao Li, Rocky K. C. Chang, Kwok-Wun Yung, Ching-Ho Chan, and Yat-Sing Poon. "An Automated Testbed for Profiling the Packet Send-time Accuracy of Embedded Devices." In: *Proc. TRIDENCOM*. June 2015.
 - [15] Ricky K. P. Mok, Weichao Li, and Rocky K. C. Chang. "Detecting Low-quality Crowdfunding Workers." In: *Proc. IWQoS (short paper)*. June 2015.
 - [16] Steven Lee and Rocky K. C. Chang. "Evaluation of Lag-Related Configurations in First-Person Shooter Games." In: *Proc. NetGames (short paper)*. Dec. 2015.
 - [17] Ricky K. P. Mok, Weichao Li, and Rocky K. C. Chang. "Improving the packet send-time accuracy in embedded devices." In: *Proc. PAM*. Mar. 2015.
 - [18] Weichao Li, Ricky K. P. Mok, Daoyuan Wu, and Rocky K. C. Chang. "On the Accuracy of Smartphone-based Mobile Network Measurement." In: *Proc. IEEE INFOCOM*. Apr. 2015.

- [19] Yajuan Tang, Xiapu Luo, Qing Hui, and Rocky K. C. Chang. "Modeling the Vulnerability of Feedback-control Based Internet Services to Low-rate DoS Attacks." In: *IEEE Trans. Information Forensics and Security* (Mar. 2014).
- [20] Daoyuan Wu and Rocky K. C. Chang. "Analyzing Android Browser Apps for file:// Vulnerabilities." In: *Proc. Information Security Conf.* Oct. 2014.
- [21] Peng Zhou, Xiapu Luo, and Rocky K. C. Chang. "Inference Attacks against Trust-based Onion Routing: Trust Degree to the Rescue." In: *Computer & Security* (Nov. 2013).
- [22] Yujing Liu, Xiapu Luo, Rocky K. C. Chang, and Jinshu Su. "Characterizing Inter-domain Rerouting by Betweenness Centrality after Disruptive Events." In: *IEEE J. Sel. Areas in Communications (SI on Network Science)* (June 2013).
- [23] Ang Chen, Edmond W. W. Chan, Xiapu Luo, Waiting W. T. Fok, and Rocky K. C. Chang. "An Efficient Approach to Multi-level Route Analytics." In: *Proc. IFIP/IEEE IM.* May 2013.
- [24] Weichao Li, Ricky K. P. Mok, Rocky K. C. Chang, and Waiting W. T. Fok. "Appraising the Delay Accuracy in Browser-based Network Measurement." In: *Proc. ACM/USENIX IMC (short paper).* Oct. 2013.
- [25] Waiting W. T. Fok, Xiapu Luo, Ricky Mok, Weichao Li, Yujing Liu, Edmond W. W. Chan, and Rocky K. C. Chang. "MonoScope: Automated Network Faults Diagnosis Based on Active Measurements." In: *Proc. IFIP/IEEE IM.* May 2013.
- [26] Peng Zhou, Xiapu Luo, Ang Chen, and Rocky K. C. Chang. "SGor: Trust Graph based Onion Routing." In: *Computer Networks* (Oct. 2013).
- [27] Yujing Liu, Xiapu Luo, Rocky K. C. Chang, and Jinshu Su. "Characterizing Inter-domain Rerouting after Japan Earthquake." In: *Proc. IFIP NETWORKING.* May 2012.
- [28] Peng Zhou, Xiapu Luo, and Rocky K. C. Chang. "More Anonymity Through Trust Degree in Trust-based Onion Routing." In: *Proc. SecureComm.* Sept. 2012.
- [29] Ricky K. P. Mok, Xiapu Luo, Edmond W. W. Chan, and Rocky K. C. Chang. "QDASH: A QoE-Aware DASH System." In: *Proc. ACM Multimedia Systems Conf.* Feb. 2012.
- [30] Xiapu Luo, Edmond W. W. Chan, Peng Zhou, and Rocky K. C. Chang. "Robust Network Covert Communications Based on TCP and Enumerative Combinatorics." In: *IEEE Trans. Dependable and Secure Computing* 9 (Nov. 2012), pp. 890–902.
- [31] Xiapu Luo, Peng Zhou, Edmond W. W. Chan, Rocky K. C. Chang, and Wenke Lee. "A Combinatorial Approach to Network Covert Communications with Applications in Web Leaks." In: *Proc. IEEE/IFIP DSN.* June 2011.
- [32] Yi Xie, Xiapu Luo, and Rocky K. C. Chang. "Centralizing the Power Saving Mode for 802.11 Infrastructure Networks." In: *Energy Technology and Management.* Ed. by Tauseef Aized. InTechOpen, 2011. Chap. 1, pp. 1–24.
- [33] Xiapu Luo, Peng Zhou, Junjie Zhang, Roberto Perdisci, Wenke Lee, and Rocky K. C. Chang. "Exposing Invisible Timing-based Traffic Watermarks with BACKLIT." In: *Proc. ACSAC.* Dec. 2011.
- [34] Xiapu Luo, Peng Zhou, Edmond W. W. Chan, Wenke Lee, Rocky K. C. Chang, and Roberto Perdisci. "HTTPOS: Sealing Information Leaks with Browser-side Obfuscation of Encrypted Flows." In: *Proc. Network and Distributed System Security Symp.* Feb. 2011.

- [35] Ricky K. P. Mok, Edmond W. W. Chan, Xiapu Luo, and Rocky K. C. Chang. "Inferring the QoE of HTTP Video Streaming from User-Viewing Activities." In: *Proc. ACM SIGCOMM Workshop on Measurements Up the Stack (W-MUST)*. Aug. 2011.
- [36] Ricky K. P. Mok, Edmond W. W. Chan, and Rocky K. C. Chang. "Measuring the quality of experience of HTTP video streaming." In: *Proc. IEEE/IFIP IM (Pre-Conf Session)*. May 2011.
- [37] Edmond W. W. Chan, Xiapu Luo, Waiting W. T. Fok, Weichao Li, and Rocky K. C. Chang. "Non-cooperative Diagnosis of Submarine Cable Faults." In: *Proc. PAM*. Mar. 2011.
- [38] Weichao Li, Waiting W. T. Fok, Edmond W. W. Chan, Xiapu Luo, and Rocky K. C. Chang. "Planetopus: A System for Facilitating Collaborative Network Monitoring." In: *Proc. IEEE/IFIP IM (Application Session)*. May 2011.
- [39] Edmond W. W. Chan, Ang Chen, Xiapu Luo, Ricky K. P. Mok, Weichao Li, and Rocky K. C. Chang. "TRIO: Measuring Asymmetric Capacity with Three Minimum Round-Trip Times." In: *Proc. ACM CoNEXT Conf.* Dec. 2011.
- [40] Edmond W. W. Chan, Xiapu Luo, Weichao Li, Waiting W. T. Fok, and Rocky K. C. Chang. "Measurement of Loss Pairs in Network Paths." In: *Proc. ACM/USENIX IMC*. Nov. 2010.
- [41] Rocky K. C. Chang, Waiting W. T. Fok, Weichao Li, Edmond W. W. Chan, and Xiapu Luo. "Neighbor-Cooperative Measurement of Network Path Quality." In: *Proc. IEEE GLOBECOM*. Dec. 2010.
- [42] Edmond W. W. Chan, Xiapu Luo, and Rocky K. C. Chang. "A Minimum-Delay- Difference Method for Mitigating Cross-Traffic Impact on Capacity Measurement." In: *Proc. ACM CoNEXT*. Dec. 2009.
- [43] Xiapu Luo, Edmond W. W. Chan, and Rocky K. C. Chang. "CLACK: A Network Covert Channel Based on Partial Acknowledgment Encoding." In: *Proc. IEEE ICC*. June 2009.
- [44] Yi Xie, Xiapu Luo, and Rocky K. C. Chang. "Centralized PSM: An AP-centric Power Saving Mode for 802.11 Infrastructure Networks." In: *Proc. IEEE Sarnoff Symp.* Mar. 2009.
- [45] Xiapu Luo, Edmond W. W. Chan, and Rocky K. C. Chang. "Design and Implementation of TCP Data Probes for Reliable and Metric-rich Network Path Monitoring." In: *Proc. USENIX Annual Tech. Conf.* June 2009.
- [46] Yajuan Tang, Xiapu Luo, and Rocky K. C. Chang. "On Generalized Low-Rate Denial-of-Quality Attack Against Internet Services." In: *Proc. IEEE IWQoS (short paper)*. July 2009.
- [47] Rocky K. C. Chang, Edmond W. W. Chan, Waiting W. T. Fok, and Xiapu Luo. "Sampling TCP Data-Path Quality with TCP Data Probes." In: *Proc. PFLDNet*. May 2009.
- [48] Xiapu Luo, Edmond W. W. Chan, and Rocky K. C. Chang. "TCP Covert Timing Channels: Design and Detection." In: *Proc. IEEE/IFIP DSN*. June 2008.
- [49] Yajuan Tang, Xiapu Luo, and Rocky K. C. Chang. "Protecting Internet Services from Low-Rate DoS Attacks." In: *Critical Infrastructure Protection*. Ed. by E. Goetz and S. Shenoi. Springer, Nov. 2007. Chap. 18, pp. 1–24.
- [50] Samantha Lo, Rocky K. C. Chang, and Lorenzo Colitti. "An Active Approach to Measuring Routing Dynamics Induced by Autonomous Systems." In: *Proc. ACM Workshop on Experimental Computer Science*. June 2007.

- [51] Xiapu Luo, Edmond W. W. Chan, and Rocky K. C. Chang. "Cloak: A Ten-fold Way for Reliable Covert Communications." In: *Proc. ESORICS*. Sept. 2007.
- [52] Xiapu Luo, Edmond W. W. Chan, and Rocky K. C. Chang. "Crafting Web Counters into Covert Channels." In: *Proc. The 22nd TC-11 IFIP Intl. Information Security Conf.* May 2007.
- [53] Hui Wang, Dah-Ming Chiu, John Lui, and Rocky K. C. Chang. "Inter-AS Inbound Traffic Engineering via ASPP." In: *IEEE Trans. Network and Service Management* 4 (June 2007), pp. 62–70.
- [54] Samantha Lo and Rocky K. C. Chang. "Measuring the Effects of Route Prepending for Stub Autonomous Systems." In: *Proc. ICC Workshop on Traffic Engineering in Next Generation IP Networks*. June 2007.
- [55] Jun Gao, Guangmin Hu, Xingmiao Yao, and Rocky K. C. Chang. "Anomaly Detection of Network Traffic based on Wavelet Packet." In: *Proc. IEEE Asia-Pacific Conf. Communications*. Aug. 2006.
- [56] Yi Xie and Rocky K. C. Chang. "Optimizing Energy Efficiency in Polling-Based Wireless Networks with Stability Constraints." In: *Proc. IEEE GLOBECOM*. 2006.
- [57] Sum Lam, Rocky K. C. Chang, and Yi Xie. "Relative stability analysis of multiple queues." In: *Proc. VALUETOOLS*. 2006.
- [58] Yi Xie and Rocky K. C. Chang. "Stability-Constrained Optimization for Energy Efficiency in Polling-Based Wireless Networks." In: *Proc. VALUETOOLS*. 2006.
- [59] Xiapu Luo, Edmond W. W. Chan, and Rocky K. C. Chang. "Vanguard: A New Detection Scheme for a Class of TCP-targeted Denial-of-service Attacks." In: *Proc. IEEE/IFIP NOMS*. Apr. 2006.
- [60] Rocky K. C. Chang, Ho Y. Chan, and Adam W. Yeung. "Deadlock-Free TCP Over High-Speed Internet." In: *Deadlock Resolution in Computer-Integrated Systems*. Ed. by Meng Chu Zhou and Maria Pia Fanti. CRC Press, Dec. 2005. Chap. 15.
- [61] Hui Wang, Rocky K. C. Chang, Dah-Ming Chiu, and John C. S. Lui. "Characterizing the Performance and Stability Issues of the AS Path Prepending Method: Taxonomy, Measurement Study and Analysis." In: *Proc. ACM SIGCOMM Asia Conf.* Apr. 2005.
- [62] Guangmin Hu and Rocky K. C. Chang. "Forwarding State Scalability-Aware Multicast Routing." In: *Proc. IEEE ISCC*. June 2005.
- [63] Rocky K. C. Chang and Michael Lo. "Inbound Traffic Engineering for Multihomed ASs Using AS Path Prepending." In: *IEEE Network* (Mar. 2005), pp. 18–25.
- [64] Xiapu Luo and Rocky K. C. Chang. "Novel Approaches to End-to-End Packet Reordering Measurement." In: *Proc. ACM/USENIX Conf. Internet Measurement*. Oct. 2005.
- [65] Xiapu Luo and Rocky K. C. Chang. "On a New Class of Pulsing Denial-of-Service Attacks and the Defense." In: *Proc. Network and Distributed System Security Symp. (NDSS)*. Feb. 2005.
- [66] Xiapu Luo and Rocky K. C. Chang. "Optimizing the Pulsing Denial-of-Service Attacks." In: *Proc. IEEE/IFIP DSN*. June 2005.
- [67] Xiapu Luo, Rocky K. C. Chang, and Edmond W. W. Chan. "Performance Analysis of TCP/AQM Under Denial-of-Service Attacks." In: *Proc. IEEE MASCOTS*. Sept. 2005.

- [68] Hang T. Chan and Rocky K. C. Chang. "Strifeshadow Fantasy: A Massive Multi-Player Online Game." In: *Proc. IEEE Consumer Commun. and Networking Conf.* Jan. 2004.
- [69] Wing W. Y. Ng, Rocky K. C. Chang, and Daniel S. Yeung. "Dimensionality Reduction for Denial of Service Detection Problems Using RBFNN Output Sensitivity." In: *Proc. IEEE Intl. Conf. Machine Learning and Cybernetics*. Nov. 2003.
- [70] Rocky K. C. Chang and Sum Lam. "Per-Queue Stability Analysis of a Dynamic Quota Sharing Scheme for Wireless Networks." In: *ACM Performance Evaluation Review (Proc. Workshop on Mathematical Performance Modeling and Analysis)*. June 2003.
- [71] Rocky K. C. Chang. "Defending Against Flooding-Based, Distributed Denial-of-Service Attacks: A Tutorial." In: *IEEE Communications Magazine (A Special Issue on Telecommunication Network Security)* 40.10 (2002), pp. 42–51.
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Posters.....

- Dec. 2015:** Daoyuan Wu, Weichao Li, Rocky K. C. Chang, and Debin Gao, "MopEye: Monitoring Per-app Network Performance with Zero Measurement Traffic," ACM CoNEXT poster.
- Aug. 2014:** Ricky K. P. Mok, Weichao Li, and Rocky K. C. Chang, "A User Behavior Based Cheat Detection Mechanism for Crowdstesting," ACM SIGCOMM Conf. poster and demo.
- Aug. 2013:** Lei Xue, Ricky K. P. Mok, and Rocky K. C. Chang, "OMware: An Open Measurement Ware for Stable Residential Broadband Measurement," ACM SIGCOMM Conf. poster and demo.
- Apr. 2013:** Ricky K. P. Mok, Weichao Li, and Rocky K. C. Chang, "IRate: Initial Video Bitrate Selection System for HTTP Streaming," USENIX NSDI poster and demo.
- Dec. 2011:** Ang Chen, Edmond W. W. Chan, Xiapu Luo, and Rocky K. C. Chang, "An Integrative Multi-level Analysis of Internet Path Changes," ACM CoNEXT Conf. poster.
- Oct. 2010:** Rocky K. C. Chang, Edmond W. W. Chan, Weichao Li, Waiting W. T. Fok, and Xiapu Luo, "Could Ash Cloud or Deep-Sea Current Overwhelm the Internet?" USENIX HotDep Workshop poster.

Nov. 2005: Samantha Lo and Rocky K. C. Chang, "Active measurement of the AS path prepending method," IEEE ICNP Conf. poster.

Aug. 2004: Steve Poon and Rocky K. C. Chang, "Robust Forwarding in Structured Peer-to-peer Overlay Networks," ACM SIGCOMM poster.

May 2003: Rocky K. C. Chang, "A Multicast-on-large-demand Approach to the Flash Crowd Problem," Twelfth Intl. WWW Conf. poster.

Invited Presentations

Jul. 2018: "Toward Scalable and Reliable Campus Wi-Fi Performance Monitoring," Huawei's Future Network Theory Laboratory, HK.

Apr. 2018: "Quality of Presentation," Research Forum, South China University of Technology.

Mar. 2018: "How to Crowdsource Mobile Users' Network Experience at Scale and with High Accuracy?" Department of Computer Science and Engineering, HKUST, HK.

Oct. 2016: "Crowdsourcing Mobile Users' Per-App Network Performance," University of Technology Sydney, Australia.

Apr. 2015: "Measurement-Driven Network Analytics," HKNOG 17, HK.

Apr. 2015: "Making Network Measurement Accurate Everywhere," Xiamen University, PRC.

Dec. 2013: "Quality of Presentation," COMP's Annual Research Day, Computing, PolyU.

Sep. 2013: "Internet Intelligence," HOTPICKS: Talk Series for Alumni, PolyU.

Jul. 2013: "Network Measurement Research in a Complex Networked World," Internet Conference of China, PRC.

Jul. 2013: "My Research Students and I," School of Computer, National University of Defense Technology, PRC.

Jul. 2013: "Measuring Internet Performance: Challenges and Opportunities," Xiamen University, PRC.

Nov. 2012: "Towards the Science of Network Measurement," Network Research Center, Tsinghua University, PRC.

Jul. 2012: "Measuring and Understanding Internet Performance: A Personal View," Graduate Forum, Sun Yat-sen University, PRC.

Jan. 2011: "Non-cooperative Diagnosis of Submarine Cable Faults," Department of Computer Science and Engineering, Behang University, PRC.

Jan. 2011: "Measurement of Loss Pairs in Network Paths," Institute of Computing Technology, Chinese Academy of Sciences, PRC.

Jan. 2011: "Active Measurement of Data-Path Quality in a Non-cooperative Internet," Tsinghua University, PRC.

Oct. 2010: "Active Measurement of Data-Path Quality in a Non-cooperative Internet," Simon Fraser University, Canada.

Dec. 2009: "Recent Advances in Active Internet Path Measurement," Sichuan University, PRC.

Jun. 2010: "Recent Advances in Active Internet Path Measurement," National University of Defense Technology, PRC.

Jun. 2009: "A Tutorial on OneProbe, JUCC Network Taskforce Seminar," PolyU, HK.

Jul. 2007: "An Active Approach to Measuring Routing Dynamics Induced by Autonomous Systems,"

Nanyang Technological University, Singapore.

Other Conference Presentations (Refereed Based on Slides).....

31 Mar. 2015: "Improving the accuracy of non-cooperative active measurement," AIMS-7, USA.

11 Feb. 2011: "OneProbe: Measuring Network Path Quality with TCP data-packet pairs," AIMS-3, USA.

Feb. 2010: "Measuring the Performance and Fairness of a Shared Network Infrastructure," APRICOT, Malaysia.

Feb. 2009: "Lessons Learnt from the Beijing Olympic Games Website Measurement," APRICOT, The Philippines.

Sep. 2008: "Lessons Learnt from the Beijing Olympic Games Website Measurement," APNOMS, PRC.

Feb. 2008: "An Active Approach to Measuring Routing Dynamics Induced by Autonomous Systems," APRICOT, Taiwan.

Research Student Supervisions (as Chief Supervisor).....

2020: Dr. Peter Membrey (EngD)

2018: Steven Lee (MPhil)

2017: Dr. Weichao Li (PhD)

2016: Dr. Ricky Mok (PhD)

2014: Dr. Peng Zhou (PhD)

2014: Dr. Daoyuan Wu (MPhil)—Received PhD from Singapore Management University later on

2010: Dr. Edmond W. W. Chan (PhD)

2009: Dr. Sam Lam (PhD)

2008: Dr. Grace Xie (PhD)

2008: Dr. Samantha Lo (MPhil)—Received PhD from Georgia Tech later on

2007: Dr. Xiapu Luo (PhD)

2000: Filli Cheng (MPhil)

1997: Dr. Hong Y. Wang (PhD, Tianjin Univ.)

1997: Sam Lam (MPhil)

1997: Fai Siu (MPhil)

1997: Simon Wong (MPhil)

Undergraduate Research Opportunities.....

2025: CS junior students Ben Kusters and Joshua Park participating in an NSF research project with UCSD.

2024–2025: CS senior students Daniel Park and Hye Lee, and CS Sophomore Eduard Dehelean participating in a research project with UCSD.

2024: CS senior students Anwesha Pradhananga and Aishwarya Joshi participated in an NSF research project with UCSD.

Services and Professional Activities

Memberships.....

1997–2018: Member, ACM

1987–2018: Member, IEEE

1987–1997: Member, Institute for Operations Research and the Management Sciences

Editorials.....

2026: Guest editor (with Ricky Mok), Advanced Technologies for Enhancing Quality of Experience (QoE), **J. Applied Sciences**, MDPI

2013: Editor (with Matthew Roughan), Proc. 14th Passive and Active Measurement Conf., **Springer's Lecture Notes in Computer Science**, Volume 7799

Since 2009: Associate Editor, **International Journal of Network Management**

2009: Guest Editor (with Chin-Tser Huang and Polly Huang), special issue on Signal Processing Applications in Network Intrusion Detection Systems, **EURASIP Journal on Applied Signal Processing**

Conference Organizations (Selected).....

2016–2017, 2019–2020: TPC Member, ACM Internet Measurement Conf. (IMC)

2016: Co-Chair, CoNEXT Student Workshop

2015–2017: TPC Member, IEEE INFOCOM Conf.

2013–2020: Steering Committee Member, Passive and Active Measurement (PAM) Conference

2013: General Chair, 14th PAM Conference

Other Professional Activities.....

2008–2009: Representative in IEEE ICC for the IEEE Technical Committee on Network Operations and Management

2004–2005: Member, Areas of Excellence on Information Technology (Network Security), HK

2007: Member, HK Focus Group for IEEE ComSoc WCET Certification Program

1999: Organizing Committee Member, Forum on Linux and Open Source, HK

1992: Internal Training Workshop on RESQME packages at IBM, NY

Strategic Planning and Advisory.....

2017–2019: Elected Staffing Committee, Computing, PolyU

2013–2018: Member, Management Committee, Computing, PolyU

1999–2000: Program Restructuring and Development Leader, Computing, PolyU

1996–2000: Member, Advisory Committee, Computing, PolyU

1996–1998: Acting Resource and Support Coordinator, Computing, PolyU

Academic Management.....

2023–2024: Member, Curriculum Committee, CS, Calvin

2013–2018: Member, Teaching and Learning Committee, Computing, PolyU

2013–2016: Member, Challengers Program Subcommittee, Computing, PolyU

2013–2015: Academic Advising Coordinator, Computing, PolyU

1999–2013: Industrial Placement Officer, Computing, PolyU

1998–1999: Examination Officer, Computing, PolyU

1997–1999: Final Year Project Coordinator, BSc (Hons) in Information Technology, Computing, PolyU

Professional Training

2014: Summer Course on Internet Protocols, Office of the Communications Authority, HK

1995: Two-day Internal Training on TCP/IP Networks, Digital Equip. Corp./HK

Other Services

2021–2023: Advisor, Chung Yuan Christian University's Student Fellowship, CYCU

2008–2013: Warden, LiZhi Hall, Student Hall of Residence, Hung Hom, PolyU

2001: Ad Hoc Committee Participant, S4-5 Computer and IT Curriculum Framework, HK

1999–2001: Member, Subject Committee of HK Examinations Authority on CE Computer Studies, HK

1994–1995: Elected Nonhead Committee Representative, Computing, HK